

Q12

Since no energy is lost to the surroundings (insulated container),

Initial total KE of molecules in both gases = Final total KE of molecules of both gases (at common temperature)

$(3/2) \mu_1 R T_1 + (3/2) \mu_2 R T_2 = (3/2) (\mu_1 + \mu_2) R T$ where μ refers to number of moles of gas.

Hence $\mu_1 T_1 + \mu_2 T_2 = (\mu_1 + \mu_2) T$

$3.0 \times 400 + 5.0 \times 200 = (3.0 + 5.0) T$ or $T = 275 \text{ K}$

Ans: B